

Lightweight Tuberculosis Screening Pipeline for Community-based Active Case Finding

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Abstract—Community-based active case finding (ACF) for tuberculosis (TB) screens at-risk populations outside health facilities, often using computer-aided detection (CAD) software. ACF is often conducted at remote sites with unreliable network connectivity, on laptops with limited computational power, and with no on-site radiologist to interpret chest X-rays. We present an open-source TB screening pipeline for adult community ACF that runs offline on a commodity laptop CPU and outputs a report for non-radiologist readers. The pipeline runs end-to-end in about 370 milliseconds on the evaluated CPU. Because the pipeline is open-source, it is cheaper compared to the per-study license fees and hardware costs of commercially available TB CAD systems. It further detects both active and latent TB, the latter underdetected in prior work. We evaluate ten classifiers and four detectors for both performance and computational cost, testing the classifiers on three TB chest X-ray datasets (TBX11K, Montgomery, and Shenzhen) against the World Health Organization (WHO) targets for sensitivity and specificity in TB CAD. Although the classifiers meet the WHO targets on in-distribution data, performance collapses zero-shot on out-of-distribution data and recovers only slightly after calibration, consistent with WHO guidance that CAD systems be recalibrated for each community site. Because public chest X-ray datasets hold few TB-specific samples, further work remains to train and evaluate the pipeline on data spanning multiple demographics from ACF sites. Further work also remains to measure latency on field hardware, detect specific radiographic findings for a full radiologist report, and assess the generated reports through radiologist and non-radiologist reader studies.

Keywords—Tuberculosis, Computer-Aided Detection, Chest X-ray, Medical Imaging, Community-based Active Case Finding

I. INTRODUCTION

Tuberculosis (TB) causes more deaths than any other infectious disease worldwide. It is caused by airborne *Mycobacterium tuberculosis* and typically affects the lungs, presenting as pulmonary TB [1], [3]. Pulmonary TB is typically diagnosed by assessing symptom history, chest radiography, and bacteriological tests. In facility screening, a symptomatic patient is interviewed by a clinician in a healthcare facility for symptoms. The clinician may request a chest X-ray (CXR) for the patient, which a radiologist interprets for abnormalities indicative of TB and writes in a radiographic report [7]. A patient whose symptoms or report indicate TB has presumptive TB. Bacteriological confirmation is done through smear microscopy, culture, or rapid molecular tests. Patients with bacteriologically confirmed TB then begin treatment [1].

In community-based active case finding (ACF), teams screen at-risk populations in community sites rather than waiting for symptomatic patients to present at a healthcare facility [2], [7]. Figure 1 shows an ACF workflow where TB screening and triage is conducted on the community site.

A. Computer-aided Detection of Tuberculosis

Computer-aided detection (CAD) systems are widely used in both facility screening and community-based ACF for TB. In 2025, the World Health Organization (WHO) 2025 approved six commercial products as meeting its TB screening performance standards [7]. CAD assistance has been shown to improve radiologist performance and allow non-radiologist performance to match radiologists [9], and several reviews report CAD accuracy approaching WHO targets across demographics [8], [34], [35], [14].

WHO endorsed CAD as an alternative to human readers for patients aged 15 and older [7]. Because CAD replaces the human reader, ACF field teams commonly do not include a radiologist [21], [22]. Despite WHO endorsement, however, clinicians across countries still value CAD as a complement to, rather than a replacement for, expert reading [33].

B. Challenges in CAD Integration in Community-based ACF

While TB CAD systems are highly accurate, CAD workflow integration, local validation, and human factors are relatively under-studied [10], [8]. ACF, in particular, faces barriers in connectivity, compute, and interpretation:

Connectivity and compute are constrained in community sites. ACF is often conducted in remote sites where connectivity is unstable and CAD software runs on laptops [21]. Several implementation studies document how these constraints shape field deployment [21], [31], [32].

CAD overlays are designed for radiologist readers, whereas ACF teams often do not have radiologists. We review the six TB CAD products approved by WHO in 2025 [7]: qXR v4.0 [15], CAD4TB v7 [16], Lunit INSIGHT CXR v3.1 [17], DrAid v2.4.4–6 [18], Genki Edge v3.4.2 [19], and InferRead DR Chest v1.0.1.1 [20]. All products output probability scores for abnormalities with visual localization through bounding boxes, outlines, or heatmaps. However, these are primarily designed for radiologists, rather than for non-radiologist readers in ACF settings. WHO further notes that the average reader in resource-limited settings is likely less proficient than the expert radiologists in CAD evaluation [7]. Despite CXR interpretation falling outside the radiographer's scope of practice, the radiographer in ACF decides whether to refer a patient for confirmatory testing using the patient's symptoms and the CAD output [22], [21]. CAD overlays also overlap poorly with expert-identified regions and omit the contextual reasoning used in diagnosis [27], hence textual descriptions have been proposed as a clinically meaningful complement to CAD overlays [38], [36], [28], [11], [37].

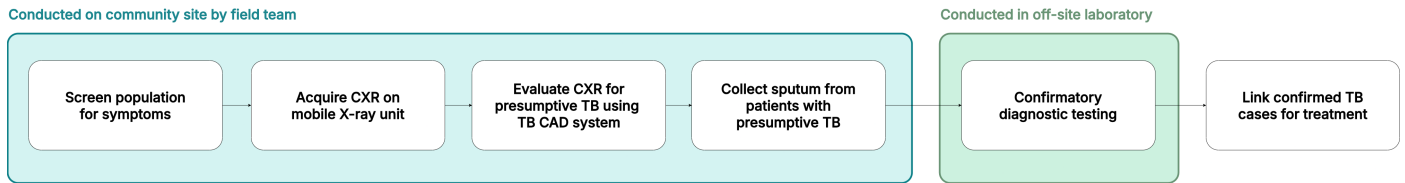


Fig. 1. A sample community-based ACF workflow for TB screening. The CAD system in ACF is often used by a radiographer rather than a radiologist.

C. Related Work

TB CAD models with lightweight memory and compute have been explored in previous studies, from LightTBNNet at about 1.5M parameters [49] to lightweight hybrid networks [50], building on edge-deployed CXR models for related conditions [25], [26]. Due to compute constraints in ACF, some existing commercial CAD systems require a dedicated CAD box or separate CAD laptop to support offline screening [21].

Related literature has also explored models that generate natural-language descriptions of findings. In a research prototype, Siemens Healthineers paired a combined classification-localization model [12] with a domain-adapted LLM to generate CXR findings [11]. General medical vision-language models such as LLaVA-Med [13] and CXR-LLaVA [29] demonstrate image-conditioned radiology text generation [30]. However, prior work remains in general chest X-ray detection, with few known to be specific to TB. Of the few TB-specific studies, generative models are used [66] that carry hallucination risk and heavier inference than a screening pipeline that must run offline on a commodity CPU.

Prior TB CAD systems validated for ACF and non-radiologist screening, such as CAD4TB [16], qXR [15], and Lunit INSIGHT CXR [17], are closed-source commercial products with per-study license and imaging-hardware costs and output specific radiographic findings for radiologist readers. To our knowledge, no prior work develops an open-source TB CAD pipeline that runs offline on a laptop CPU with report generation designed primarily for non-radiologist readers.

D. Contributions

This study presents a lightweight TB screening pipeline for community ACF where there is no on-site radiologist, no GPU, and unreliable connectivity. We present the following:

- A screening pipeline that runs offline on a commodity laptop CPU, completing a study in about 370 ms for community ACF constraints.
- A template-based natural-language report for non-radiologist ACF readers to base referral decisions.
- Open-sourced TB CAD pipeline as a cheaper alternative to per-study licensing and hardware costs of commercial systems.
- Evaluation of four detectors for both computational cost and detection of active & latent TB, where latent TB is underdetected in prior work.
- Evaluation of ten classifiers against WHO TB CAD targets, measured in-distribution on TBX11K and out-of-distribution on Montgomery and Shenzhen datasets.

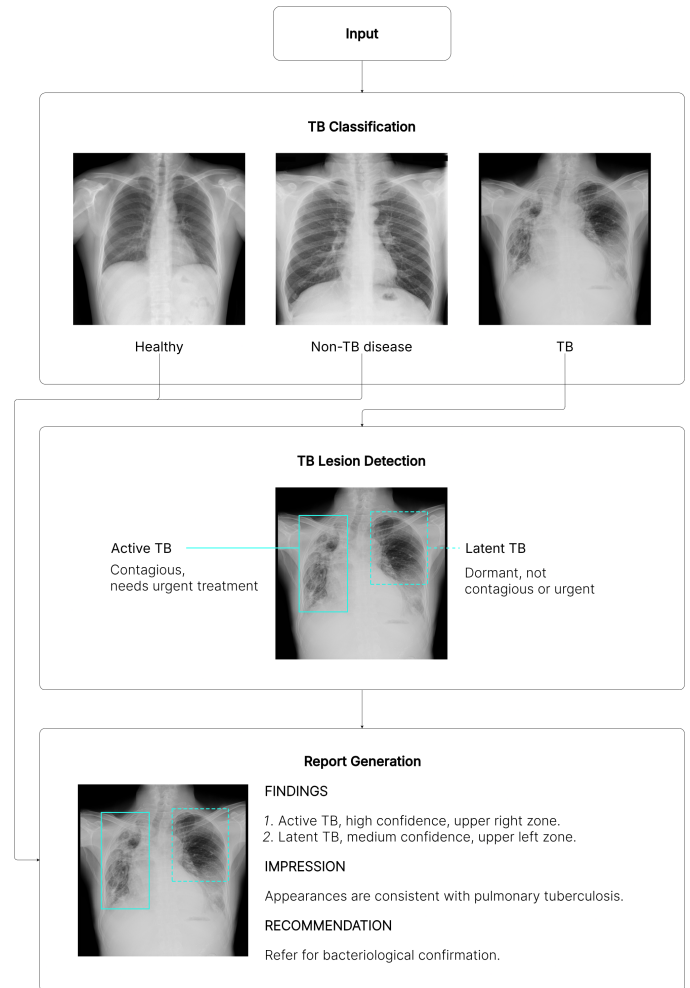


Fig. 2. Overview of the TB CAD pipeline.

II. SYSTEM OVERVIEW

An overview of the TB CAD pipeline is shown in Figure 2. A CXR (in DICOM or standard image format) is ingested from the screening unit's X-ray or uploaded directly, then scored by the TB classification module. For images flagged as TB, the object detection module identifies active and latent TB regions. The classification and detection outputs are passed to the report generation module, which outputs a natural-language report of the detected TB regions in a format similar to radiologic convention. Report findings consist of active TB and latent TB lesions rather than specific radiographic findings typical to radiologic reports because the report is designed for non-radiologist readers. All three stages are lightweight for ACF screening where no radiologist and often no GPU is present.

III. TUBERCULOSIS CLASSIFICATION

The TB classification module flags each CXR as either healthy, sick non-TB, or TB as shown in Figure 2.

A. Dataset

Classification and object detection models are trained on a simplified redistribution¹ of TBX11K [39], a public CXR dataset with image-level labels (*healthy*, *sick non-TB*, *TB*) and bounding-box-level labels (*active-TB*, *latent-TB*).

TBX11K is selected because it is the largest public TB CXR dataset with a TB-specific sample large enough to train from scratch, expert bounding-box annotations, a sick non-TB class, and a demographic and class composition that reflect real-world clinical settings [40]. Its image-level labels are established by diagnostic microbiology [40], the reference standard for TB confirmation [40], [7]. TBX11K is composed mostly of TB cases aged 20-60, predominantly among males (72% of TB X-rays), which is representative of real-world TB distribution [40].

The official TBX11K test set is withheld for an online challenge, so we use only the public trainval images. Our results are therefore not directly comparable to SymFormer [40] or other challenge entries on the private test set. The data used is made available in the Data and Code Availability section of this study.

TABLE I. TBX11K DATASET COMPOSITION (IMAGE-LEVEL LABELS) FOR CLASSIFICATION

Class	Train	Val	Test	Total
Healthy	2,660	570	570	3,800
Sick non-TB	2,660	570	570	3,800
TB	560	120	120	800
Total	5,880	1,260	1,260	8,400

Classification uses the image-level labels shown in Table I. The 8,400 public CXRs (3,800 healthy, 3,800 sick non-TB, 800 TB) are partitioned per class into train, validation, and test folds in a 70%/15%/15% ratio. All reported classification metrics are computed on the held-out test fold of 1,260 CXRs (570 healthy, 570 sick non-TB, 120 TB).

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation. TBX11K CXRs are acquired under a standardized posteroanterior protocol and are largely uniform in pose, so augmentation risks introducing distribution shift.

C. Model Architecture

We evaluate ten classifiers: six baselines (EfficientNet-B0 [44], MobileNetV3-Large [43], DenseNet-121 [42], ResNet-18, ResNet-50 [41], ConvNeXt-Tiny [45]), three custom models (DraxNet, Drax-MobileNetV3-Large, FlipR), and LightTBNNet [49], a published TB-specific lightweight network retrained under our protocol. Full architecture of custom models are listed in Appendix A.

1) *DraxNet*: DraxNet (16.5M parameters) modifies a ResNet-18 backbone so the fourth stage consists of two residual blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged. *DraxBlock* consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch, fused via stochastic-depth residuals. DraxNet, short for Dynamic Residual Attention eXchange, reserves the heavier mixer for the final, lowest-resolution stage, adding attention capacity with fewer parameters than a full attention backbone.

2) *Drax-MobileNetV3-Large*: Drax-MobileNetV3-Large (4.8M parameters) applies the DraxBlock to a parameter-efficient backbone. The MobileNetV3-Large feature extractor and head are unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage. This design tests whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) *FlipR*: FlipR (2.8M parameters) inserts a *FlipR* block between the second and third stages of a ResNet-18 truncated to three stages, with the parameter-heavy fourth stage removed. The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart, smoothed by average pooling to account for imperfect anatomical symmetry, then broadcasts a gated mask across all channels. The block is inspired by SymAttention [40], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can match heavier attention approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size 16, 512×512 input, 100 epochs, Adam optimizer at a fixed learning rate of 10^{-3} , and unweighted cross-entropy loss. No hyperparameter tuning was done. The checkpoint with the lowest validation loss is retained and evaluated on the held-out test fold.

E. Results

Classification results are reported in Table II.

No single model is lightest on both storage and compute. Lighter models show a trade-off between storage use through params and computational cost through MACs. LightTBNNet has the fewest parameters (1.3M) at the cost of higher MACs (14.1G), and models with MobileNet backbones have the fewest MACs (1.2 – 1.3G) at the cost of higher parameters (4.2 – 4.8M). Hence, the model choice depends on whether storage or compute is the tighter constraint in deployment.

Bilateral asymmetry is a strong indicator of TB. FlipR is the second smallest model by parameters (2.8M) and achieves performance within 1% of the top four baselines (LightTBNNet, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) in both accuracy and F1. FlipR's performance despite lighter parameters is consistent with the SymFormer [40] observation that bilateral asymmetry is a strong indicator of TB. Its parameter savings come from removing the parameter-heavy final stage of ResNet-18, which runs at low resolution and thus contributes little compute. FlipR roughly halves LightTBNNet's compute (7.4G) but still exceeds the MobileNet backbones (MobileNetV3-Large 1.2G, Drax-MobileNetV3-Large 1.3G).

¹<https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified>

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC _{TB}	Classification metrics (%)							Computational metrics	
		Acc	Sens	Spec	Prec	Rec	F1	Spec@90 Sens	Params	MACs
LightTbNet [49]	0.9994	98.89	94.17	99.82	98.72	97.65	98.17	99.82	1.3M	14.1G
FlipR	0.9987	98.81	94.17	99.56	98.01	97.59	97.79	99.82	2.8M	7.4G
EfficientNet-B0 [44]	0.9995	98.89	95.00	99.65	98.29	97.87	98.07	100.00	4.0M	2.1G
MobileNetV3-Large [43]	0.9995	99.21	96.67	99.82	98.97	98.54	98.75	100.00	4.2M	1.2G
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	98.23	97.81	98.02	100.00	4.8M	1.3G
DenseNet-121 [42]	0.9975	98.57	92.50	99.56	97.81	96.97	97.38	99.82	7.0M	15.0G
ResNet-18 [41]	0.9982	98.17	93.33	99.21	96.71	96.90	96.80	99.91	11.2M	9.5G
DraxNet	0.9988	98.41	95.00	99.47	97.51	97.51	97.51	100.00	16.5M	10.9G
ResNet-50 [41]	0.9978	97.70	92.50	99.47	96.95	96.33	96.63	99.47	23.5M	21.5G
ConvNeXt-Tiny [45]	0.9952	97.14	95.00	97.98	93.63	96.58	94.97	99.82	27.8M	23.4G

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB). Params is the trainable parameter count. MACs is the number of multiply-accumulate operations for one forward pass. FLOPs are twice MACs. Spec@90Sens is the specificity when sensitivity is set to 90%, which is the operating point calibration used in WHO 2025 evaluation. WHO recommends minimum 40% specificity when fixed at 90% sensitivity [7].

All models exceed WHO 2025 TB CAD targets on the TBX11K test set, but likely reflects lower task difficulty compared to the WHO test set. In the 2025 WHO TB CAD evaluation, two conditions are evaluated: 1.) specificity must meet a minimum of 40% when sensitivity is fixed to 90%, and 2.) partial AUC over the 40 to 60% specificity range is evaluated [7]. All models exceed the minimum specificity target as shown in Table II and achieve a perfect partial AUC of 1.00 over the 40 to 60% specificity range. However, WHO selected 40% specificity as this is close to expert radiologist performance on WHO's private test set, not on the TBX11K test set, hence results are indicative but not directly comparable. The WHO test set is presumed to be more difficult than the TBX11K test set. Although both are labelled based on microbiological reference standard, TBX11K is single-demographic and curated from facility screening CXRs, whereas the WHO test set is from multiple demographics with CXRs specifically from community screening. Because no known public TB dataset matches this, evaluating the classifiers on a comparable multi-demographic dataset against radiologist performance remains future work.

The high-specificity, lower-sensitivity profile is likely attributable to TBX11K rather than model architecture. All models display high specificity and lower sensitivity, consistent with results from models in related literature trained on TBX11K. Sensitivity is more clinically important than specificity in screening, because a missed TB case risks untreated transmission while a false positive triggers only unnecessary confirmatory testing. The TBX11K dataset is constructed to reward high specificity because the common failure mode on other TB datasets is over-predicting TB and achieving near-zero specificity. This is consistent with the performance of other models on the TBX11K dataset [40].

Lightweight models are preferred for TB screening on both performance and deployment grounds. The smaller models (LightTbNet, FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) match or exceed every larger model in accuracy and F1. Due to a modest TBX11K

training set, a minority TB class, and no class-imbalance handling, we hypothesize that this is because larger networks overfit the majority healthy and sick non-TB classes rather than learn discriminative TB features. Due to compute constraints in ACF, lightweight models are preferred for both performance and practical deployment.

F. Out-of-distribution Validation

To test out-of-distribution performance, we run each classifier without retraining on two public CXR sets from different countries and scanners: the Montgomery County set (138 images, 58 TB) and the Shenzhen set (662 images, 336 TB) [59]. Results are reported in Table III.

Zero-shot performance collapses on out-of-distribution data. Both Montgomery and Shenzhen datasets are binary classification tasks, hence performance is evaluated with no TB and sick non-TB labels from TBX11K as the negative class and TB as the positive class. Mean AUC_{TB} for all models falls from 0.998 within TBX11K to 0.564 across the two external sets, with several models at or below chance. Figure 3 shows this collapse for the deployed FlipR classifier. The saturated in-distribution figures therefore reflect a single curated source rather than transferable screening skill. Models should be trained on a more diverse dataset in future work and calibrated per community site as per WHO guidance [7].

Model performance on the two sets fail in opposite directions due to shifts in image appearance across sets. On Montgomery the models lose sensitivity, under-flagging TB images, while on Shenzhen the models lose specificity, over-flagging normal images. The two sets differ from the TBX11K training images in opposite ways in mean pixel intensity (Montgomery is darker at 113, Shenzhen is brighter at 157, compared to 123 for TBX11K), with a Jensen-Shannon divergence of intensity histograms from the training set of 0.30 and 0.29 respectively (Figure 6). Because the direction of failure on each set is consistent with this, the collapse is likely due to a shift in image appearance rather than an artifact of adapting the three-class model to binary.

TABLE III. AUC_{TB} OF MODELS ON OUT-OF-DISTRIBUTION DATASETS

Model	In-dist	Montgomery			Shenzhen		
		Zero-shot	AdaBN	Crop	Zero-shot	AdaBN	Crop
LightTBNNet [49]	0.999	0.470	0.571	0.432	0.394	0.375	0.430
FlipR	0.999	0.669	0.618	0.645	0.688	0.676	0.705
EfficientNet-B0 [44]	0.999	0.466	0.602	0.551	0.614	0.463	0.717
MobileNetV3-Large [43]	0.999	0.426	0.589	0.260	0.511	0.326	0.688
Drax-MobileNetV3-Large	0.999	0.571	0.725	0.384	0.475	0.434	0.569
DenseNet-121 [42]	0.997	0.511	0.537	0.552	0.495	0.545	0.594
ResNet-18 [41]	0.998	0.613	0.610	0.568	0.639	0.565	0.727
DraxNet	0.999	0.563	0.591	0.599	0.604	0.599	0.754
ResNet-50 [41]	0.998	0.636	0.625	0.635	0.637	0.612	0.785
ConvNeXt-Tiny [45]	0.995	0.697	–	0.599	0.613	–	0.687
Mean	0.998	0.562			0.567		

In-dist is the within-TBX11K AUC_{TB} from Table II. **Zero-shot** applies the model as trained. **AdaBN** recomputes batch-normalization statistics on the unlabeled target images. AdaBN is not applicable to ConvNeXt-Tiny, which uses layer normalization rather than batch normalization. **Crop** crops the image to the lungs. Bold marks the best correction (among zero-shot, AdaBN, and crop) per model and dataset.

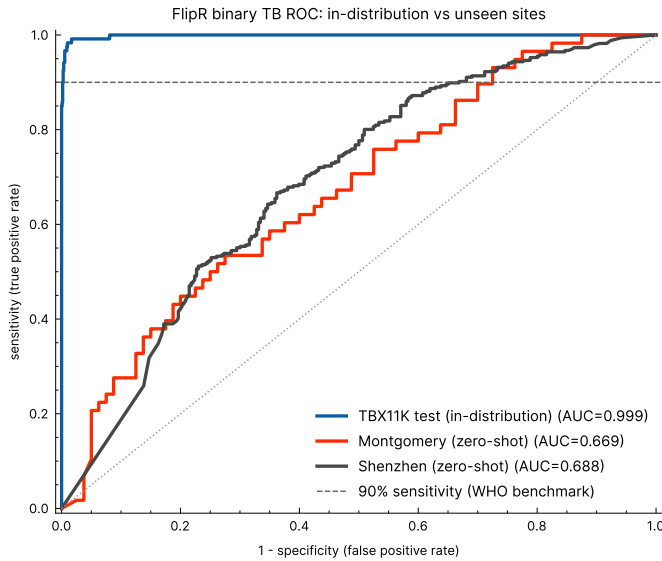


Fig. 3. Binary TB ROC for the FlipR classifier on the three datasets. The dashed line marks the 90% sensitivity operating point from WHO 2025 targets (minimum 40% specificity at 90% sensitivity) [7].

Corrections recover model performance without re-training. Real ACF CAD deployment should adapt to a new community site without relabeling, so we test two standard corrections that use only the unlabeled target images and leave all learned weights fixed. Adaptive batch normalization (AdaBN) recomputes each batch-normalization layer's running statistics on the target images [60]. Lung-field cropping restricts the input to the lungs using an off-the-shelf segmentation model [61], removing burned-in text, borders, and field-of-view differences that are known cross-dataset shortcuts [62]. The two corrections address different shift components. AdaBN recovers Montgomery, the intensity-shifted set, raising MobileNetV3-Large from 0.426 to 0.589 and Drax-MobileNetV3-Large from 0.571 to 0.725, whereas cropping

recovers Shenzhen, the framing-shifted set, raising ResNet-50 from 0.637 to 0.785 and DraxNet from 0.604 to 0.754. Neither correction helps both datasets, consistent with each addressing a distinct difference in each dataset from training data. Across the ten models, applying the better correction per set raises mean AUC_{TB} from 0.544 to 0.643, with the strongest models reaching 0.72 to 0.79. However, these figures remain below the in-distribution values and below the WHO operating targets, and thus do not establish deployment readiness. The results show instead that performance collapse is likely due to need for site calibration, which is correctable and consistent with WHO's guidance to calibrate TB CAD systems per community site [7], rather than due to a loss of learned TB features.

IV. TUBERCULOSIS LESION DETECTION

The object detection module localizes each abnormality on the CXR with a bounding box and classifies each abnormality as either an active TB or latent TB lesion.

A. Dataset

TABLE IV. TBX11K DATASET COMPOSITION (BOUNDING BOX LABELS) FOR LESION DETECTION

Annotation	Train	Test	Total
Active-TB boxes	724	248	972
Latent-TB boxes	178	61	239
Total boxes	902	309	1,211

Counts of bounding boxes. The 800 TBX11K TB images contain 1,211 boxes because a single image with TB may have multiple lesions.

Object detection uses the TBX11K bounding-box annotations, with each region labeled active TB or latent TB. Dataset composition is shown in Table IV.

Active TB denotes ongoing, transmissible infection requiring treatment. Latent TB denotes prior, inactive infection

including healed lesions that are not contagious but may reactivate. As Figure 2 shows, active TB is more visually salient and latent TB is more subtle. A CXR with an image-level label of TB may contain bounding box labels of only active-TB lesions (630 images), only latent-TB lesions (139 images), or both (30 images). This composition is consistent with real-world conditions, where active TB is more common than latent TB [39].

Differentiation between active and latent TB is relatively understudied in TB CAD. In routine reporting, a radiologist may not always state whether a lesion is suspected as active or latent TB explicitly, instead inferring it from radiographic pattern, with consolidation and cavitation suggesting active disease and fibrotic linear opacities suggesting healed disease, alongside clinical and laboratory correlation [4]. Current literature understudies whether existing commercial CAD systems can differentiate latent from active TB [21].

B. Model Architecture

We evaluate four detectors at 256×256 px input: base YOLO26-n [46], two standard nano baselines YOLOv8n [47] and YOLO11n [48], and a custom variant in which YOLO26's default feature extractor is replaced by the DraxNet backbone used in classification. All detectors are trained from random initialization under the same detection head and training protocol. The DraxNet variant retains YOLO26's head, which isolates the backbone's effect relative to base YOLO26.

C. Model Training

Object detection follows the classification setup in training all detectors from random initialization for 100 epochs at batch size 16, but uses the Ultralytics default detection optimizer (auto-selected, base learning rate 10^{-2}) rather than the Adam optimizer at learning rate 10^{-3} used for classification. The best checkpoint (selected by mAP) is retained and used for evaluation.

D. Results

Detection results are reported in Table V.

Active TB is more likely to be detected than latent TB. All detectors localize active TB several times more accurately than latent TB (roughly $3\text{--}6\times$ more accurate for active TB at mAP_{50} and $3\text{--}9\times$ more accurate at $\text{mAP}_{50\text{--}95}$). This gap reflects the TBX11K class imbalance (active-TB regions outnumber latent-TB regions roughly 4:1) and the subtle appearance of latent-TB lesions. TBX11K also reports the same pattern across all detection methods, attributing it to class imbalance [40]. This pattern is also consistent with radiologist reader performance, where experts distinguish latent TB less reliably than active TB [5]. The bias toward active TB aligns with clinical triage priorities, because active TB requires urgent treatment.

The DraxNet backbone improves active-TB localization, but by too small a margin to justify its cost. YOLO26 with the DraxNet backbone attains the best active-TB localization at 0.5865 mAP_{50} and 0.2748 $\text{mAP}_{50\text{--}95}$, ahead of the stock YOLOv8n (0.5563 and 0.2582) and YOLO11n (0.5434 and 0.2446). This performance gain comes from replacing

base YOLO26's feature extractor: DraxNet raises active-TB mAP_{50} from 0.4511 to 0.5865 (+0.1354) and $\text{mAP}_{50\text{--}95}$ from 0.2110 to 0.2748 (+0.0638) over the base YOLO26, driven almost entirely by precision ($0.74 \rightarrow 0.82$) at unchanged recall (roughly 0.26). The improvement does not extend to latent TB, where DraxNet is the weakest of the four (0.1017 mAP_{50} against 0.1175 for YOLOv8n). However, DraxNet's improvement over YOLOv8n is small, about 0.03 mAP_{50} , while its computational cost is large. The DraxNet backbone uses 19.7M parameters compared to YOLOv8n's 3.2M. The marginal active-TB gain does not justify roughly six times the parameters and compute, so we adopt YOLOv8n as the detection model, which is the strongest of the lightweight detectors and within 0.03 mAP_{50} of the DraxNet backbone's performance at a fraction of the computational cost.

Recall, not precision, is the bottleneck. Precision stays high while recall lags behind for every detector, so the detectors tend to localize detected TB correctly but miss many TB regions. Figure 4 shows recall sits well below precision and the latent-TB curve collapses at low confidence. The pattern is uniform across detectors. Aggregate recall stays near 0.26–0.28 for all four models, so no detector flags a substantial fraction of the lesions.

Localization is useful to confirm (rule-in) TB, but not to rule it out. A test rules in disease when a positive result confirms it (rule-in diagnosis) and rules out disease when a negative result excludes it (rule-out diagnosis). The high-precision, low-recall profile indicates that a flagged lesion is likely correct and warrants radiologist review, but the model cannot reliably exclude TB because most annotated lesions are missed. Localization is therefore most reliable as supporting region-level evidence for the classifier and report generation modules, rather than as a standalone detector for TB. Clinicians using the model may reliably use it for rule-in diagnosis, not for rule-out diagnosis.

V. REPORT GENERATION

The report generation module creates a report for non-radiologist readers to base referral decisions upon. In ACF the reader is a non-radiologist (often a radiographer), so the report is intended to indicate only active or latent TB, rather than full radiographic findings (ex. pleural effusions, cavitations), which are conventional for radiology reports but less intuitive for non-radiologist readers. Example outputs are shown in Figure 5.

This module follows the two-stage detector-to-text architecture of Danu et al. [11]. However, because our detector emits a small, closed vocabulary rather than full radiographic findings, we replace their proprietary detector and fine-tuned language model with the public models of this paper and template-based report generation. Use of a generative model is reserved for future work that detects specific radiographic findings. For current use, template-based report generation suffices and does not require significant memory or compute for resource-constrained ACF settings.

The module has two stages: an adapter that converts detector output into a structured record, and a template-based report generator that renders that record as text. This architecture prioritizes faithfulness to model findings, because image-conditioned vision-language model architectures such

TABLE V. TUBERCULOSIS LESION DETECTION RESULTS

Model	Active TB		Latent TB		Params	MACs
	mAP ₅₀	mAP ₅₀₋₉₅	mAP ₅₀	mAP ₅₀₋₉₅		
YOLO26 [46]	0.4511	0.2110	0.1070	0.0410	2.6M	0.5G
YOLO11n [48]	0.5434	0.2446	0.1166	0.0435	2.6M	0.5G
YOLOv8n [47]	0.5563	0.2582	0.1175	0.0444	3.2M	0.7G
YOLO26 + DraxNet	0.5865	0.2748	0.1017	0.0310	19.7M	3.2G

MACs is the number of multiply-accumulate operations for one forward pass. FLOPs are twice MACs.

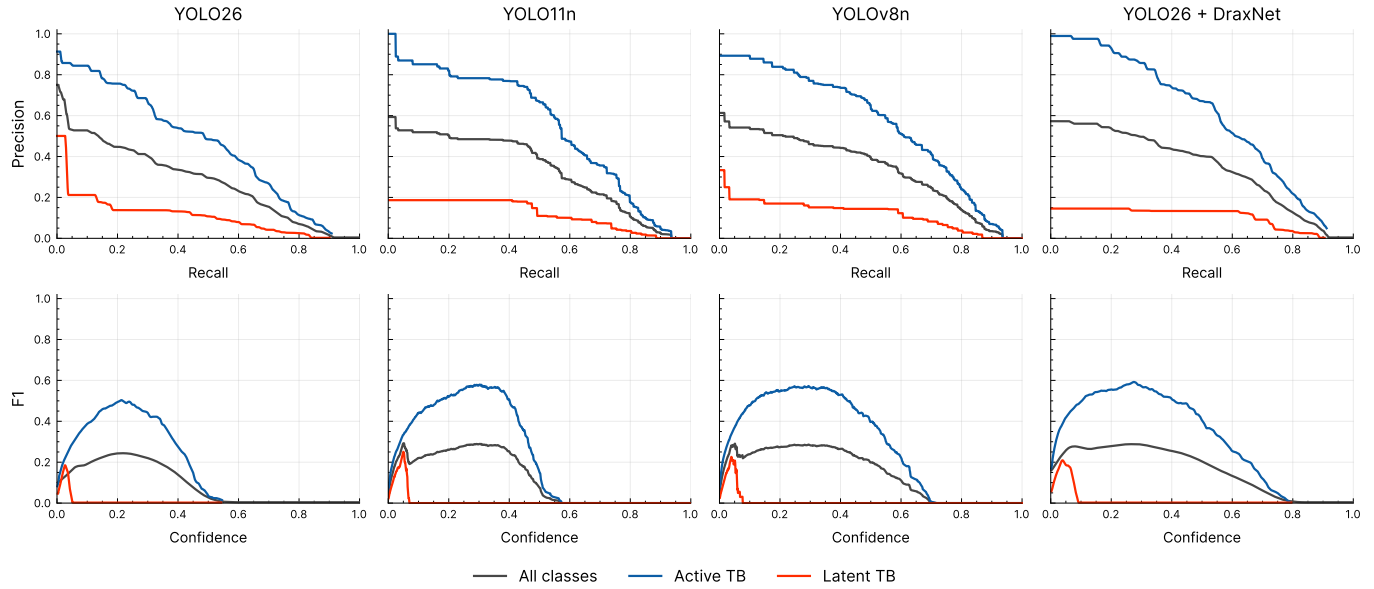


Fig. 4. Detection precision-recall (top) and F1-confidence (bottom) curves for the four detectors.

as CXR-LLaVA [29] & LLaVA-Med [13] and generative multimodal models [11] may hallucinate findings the detector never flagged.

A. Report Adapter

The adapter maps classification and detection output to a fixed schema with a small, closed vocabulary. The image-level field carries the predicted label, the argmax over *healthy*, *sick non-TB*, and *TB*, together with its class probabilities. Each detected region carries three fields: its type (*active TB* or *latent TB*), a confidence band based on the detection score (*low* at < 0.5 , *medium* at < 0.8 , or *high* at ≥ 0.8), and a location. The location is assigned by mapping the bounding-box centroid to one of nine cells in a 3×3 grid over the image (upper, middle, or lower row; left, center, or right column). An image with no detections yields an empty region list.

All images are posteroanterior (PA) frontal scans. Hence, the standard radiological laterality convention is used, wherein the patient's right side appears on the image's left side and vice versa. For example, a lesion in the image's upper left corner is reported as an upper right lesion.

B. Template-Based Report Generation

The record is rendered into a structured template in three labeled sections, mirroring the findings-to-impression structure

of a radiology report [57]: a numbered FINDINGS list with one TB region per line including lesion type, model confidence band, and lesion zone, an IMPRESSION stating the screening result in clinical wording, and a RECOMMENDATION based on the predicted class.

The report describes only the detected TB regions and does not name specific radiographic findings such as consolidation, cavitation, or fibrosis, nor cardiac, pleural, and skeletal observations. This is intended for non-radiologist readers in ACF (often radiographers), for whom chest X-ray interpretation falls outside the scope of practice and whose task is triage and referral rather than diagnosis. Specific findings that require radiological training to interpret may not be intuitive to non-radiologist readers, whereas the active-versus-latent distinction and recommendation that bears on referral urgency are what this non-radiologist reader can act on. Richer findings become useful when a radiologist receives the CXR later in a facility, which is reserved for future work.

We chose a template over a generative model after prototyping both. A language-model report generator (Phi-3.5-mini-instruct [56] run on the record from the adapter) was prototyped alongside this work, but the target is a short, fixed-format report with one line per region, which a template reproduces exactly over the closed vocabulary. The template therefore matches the generative output on faithfulness, read-

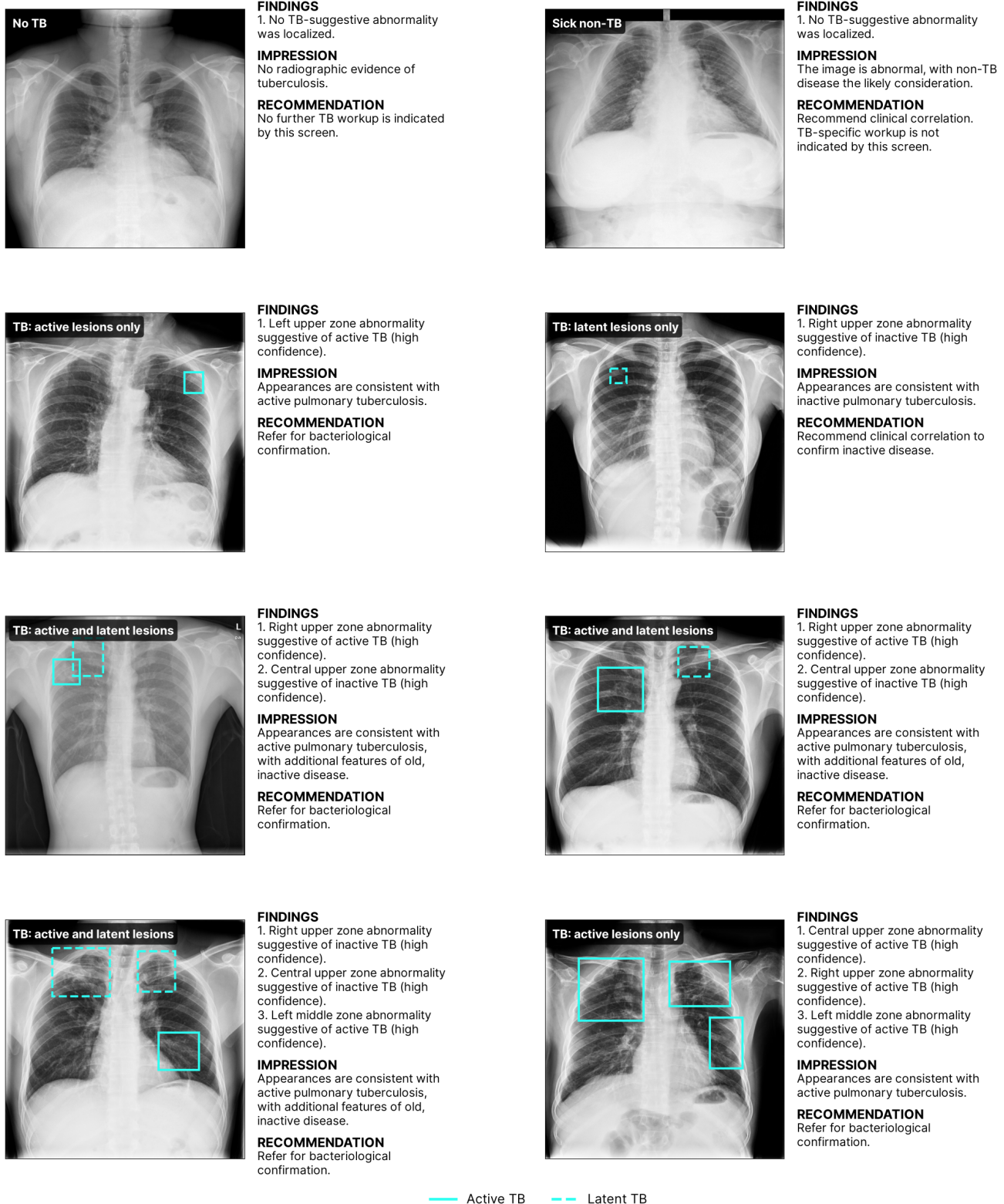


Fig. 5. Examples of generated reports. The ground-truth class is overlaid at the top-left of each CXR. Radiologic laterality is used (i.e. lesions on the left side of the image are reported as appearing on the patient's right, and vice versa).

ability, and the ordering of findings. Since the generative approach does not provide additional benefit over the template, the generative model is dropped to avoid hallucination risk

and reduce computational cost. A generative model would be needed for future work when the detector reports a richer, open set of interacting radiographic findings.

TABLE VI. ON-DEVICE CPU BENCHMARK

Model	Size (MB)	Lat. 1T (ms)	Lat. 6T (ms)	RAM (MB)
<i>Classification</i>				
LightTBNNet [49]	5.1	621	214	800
FlipR	11.2	244	76	603
EfficientNet-B0 [44]	16.3	176	100	654
MobileNetV3-Large [43]	17.0	89	58	655
Drax-MobileNetV3-Large	19.4	91	56	631
DenseNet-121 [42]	28.4	518	259	806
ResNet-18 [41]	44.8	298	91	629
DraxNet	66.0	362	114	649
ResNet-50 [41]	94.4	682	241	819
ConvNeXt-Tiny [45]	111.4	683	217	786
<i>Detection</i>				
YOLO26 [46]	5.4	110	68	620
YOLO11n [48]	5.5	120	56	608
YOLOv8n [47]	6.2	124	46	610
YOLO26 + DraxNet	39.7	420	152	727

Measured on an Intel Core i7-8750H CPU using PyTorch with float32 compute, batch 1, 512×512 input, and no GPU. **Size** is the size of the checkpoint on disk. **Lat. 1T** and **Lat. 6T** are the median single-image forward-pass latency at 1 and 6 CPU threads over 40 runs. **RAM** is the peak RAM usage during the inference process, which includes a fixed runtime of about 330 MB (360 MB with the detection library). These runtime figures are specific to this host and are not comparable across papers. To compare with related literature, see the parameters and MACs in Tables II and V.

C. Faithfulness by Design

Report generation is faithful by design, meaning the generated text does not assert findings that are not present in the structured record. This is based on the faithfulness-versus-factuality distinction for abstractive summarization [51] and data-to-text evaluation [52], under which a generation is faithful if every entity it states is supported by the source, independent of whether the source is itself correct. Faithfulness here is the absence of extrinsic hallucination. It does not by itself imply factual correctness, which is based on the accuracy of the upstream classification and detection models.

VI. ON-DEVICE DEPLOYMENT

A. Deployment Feasibility

To test on-device model performance, we benchmark each model on a commodity laptop CPU in Table VI.

Laptops are typically used to run CAD systems in community-based ACF, resulting in limited compute. CAD is also increasingly deployed offline where network connectivity is unreliable or absent [23], [21], [24]. Thus, models for community-based ACF should preferably be able to run locally on a CPU, without a dedicated GPU or a network connection.

1) *Memory Footprint*: On-disk size is primarily affected by parameter count. Our lightweight models fall at the parameter scale shown in related literature to run on CPU-only hardware. Lightweight chest X-ray networks have been deployed on edge devices and CPUs with low latency, including ResNet-18, MobileNet, and EfficientNet-B0 on an embedded board [25]

and a compact classifier on edge devices [26], and for TB specifically LightTBNNet reports rapid, low-memory prediction designed for embedded use [49]. LightTBNNet (1.3M parameters), FlipR (2.8M), EfficientNet-B0 (4.0M), MobileNetV3-Large (4.2M), and Drax-MobileNetV3-Large (4.8M), and all tested stock YOLO detectors (2.6M-3.2M) are at or below the size of networks deployed on embedded boards in [25], [26], so they should run on the same devices.

2) *Memory-Compute Tradeoff*: Tested classifiers tend to have a tradeoff between memory and compute. Although LightTBNNet carries the lowest memory footprint (1.3M parameters; 5.1MB size), it costs high compute (14.1G MACs). FlipR follows the pattern of lower memory footprint (2.8M parameters; 11.2MB size) with higher compute (7.4G MACs). Meanwhile, models with MobileNet backbones have lower compute requirements (1.2-1.3G MACs) but a higher memory footprint (4.2-4.8M parameters; 17.0 – 19.4MB size). Hence, the classifier selected for deployment should be chosen based on whether memory or compute is more constrained in the target device. This is less relevant for detectors, which uniformly have higher memory footprint and lower compute.

3) *Inference Latency*: On-device latency is affected primarily by MACs. Every classifier completes a forward pass in under 0.7 seconds on a single thread and under 0.26 seconds on six threads. The MobileNet backbones are fastest at about 90 ms on a single thread. FlipR uses fewer parameters than MobileNet backbones but takes 244 ms on a single thread. FlipR inference latency reduces significantly to 76 ms on six threads because FlipR's standard convolutions parallelize across cores more efficiently than depthwise convolutions in MobileNet backbones. Despite LightTBNNet using the fewest parameters, its high-resolution architecture gives it the third-highest single-thread latency at 621 ms and fourth-highest six-thread latency at 214 ms, similar to heavier models DraxNet, ResNet-50, and ConvNeXt-Tiny. Of the detectors, stock YOLO models have similar inference latency (110-124 ms single-thread, 46-68 ms six-thread).

4) *RAM Usage*: RAM use is primarily based on activation size. All models' peak RAM usage fall within 600-820 MB. Out of the classifiers, LightTBNNet holds the fewest parameters yet the third-highest peak memory at 800 MB, because its high-resolution feature maps dominate the footprint. FlipR, EfficientNet-B0, and models with MobileNet backbones show peak RAM usage around 603-655 MB. Stock YOLO detectors show similar peak usage around 608-620 MB.

5) *Selected Configuration*: For the abstract, conclusion, and cost analysis, we report figures for the FlipR-YOLOv8n models based on evaluation of performance and computational cost. Among the five classifiers with the lowest memory and compute cost (LightTBNNet, FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large), FlipR is selected because it achieves the highest zero-shot performance on external datasets Montgomery and Shenzhen. Among the three detectors with the lowest memory and compute cost (YOLOv8n, YOLO11n, YOLO26n), YOLOv8n is selected because it achieves the highest mAP scores for both active TB and latent TB. However, choice of deployment model for practical use should be chosen on the basis of constraints on the target device and/or community site.

B. Deployment Cost

We estimate representative costs on the selected configuration, FlipR-YOLOv8n, below. However, as deployment costs are highly variable based on implementation and deployment details, this cost analysis is intended to be an example estimate.

1) *Cost Estimate for Selected Configuration:* The selected configuration, FlipR-YOLOv8n, completes a CXR study in about 370 ms on a single thread (Table VI). A screening program of 100,000 studies per year therefore needs about 10 CPU-hours of compute per year. At a commodity cloud CPU rate near 0.045 USD per core-hour, that is under 0.50 USD per year, and on a single 300 USD offline mini-PC amortized over a three-year service life, the same workload comes to about 110 USD per year while leaving the device idle for most of the day. In community ACF, where the team already runs CAD on a laptop it owns [21], the marginal cost of an additional study is effectively only the electricity cost, because the pipeline is open source and does not require per-study license costs.

2) *Cost Estimate for Commercial TB CAD Systems:* Commercial TB CAD systems either have per-study or perpetual license costs, where prices are negotiated directly with health providers and are not published online [63]. The firmest public figures are the concessional bundle prices through the Stop TB Partnership Global Drug Facility, around 16,700 USD for a perpetual CAD4TB license with hardware, installation, and one year of support. A multi-product cost model for Pakistan reports per-screen costs from 0.19 to 2.73 USD, where perpetual-license products approach a low per-screen cost only at high throughput and volume-priced products stay near the 0.93 USD cost of a radiologist read [63]. Field programs report 13.31 USD per screen for CAD4TB in Zambia [64] and a 9,000 USD annual license alongside a 75,000 USD imaging device for qXR in Nigeria [65]. However, these figures are highly dependent on the context of screening (i.e. base year, country-based pricing, currency exchange rates), so we report these figures as context rather than as a direct comparison against the estimated compute cost for FlipR-YOLOv8n above.

3) *Primary Factors for Deployment Costs:* Deployment cost is mainly driven by costs of TB CAD licensing, imaging hardware, and program logistics. As the pipeline is open source, it requires no per-study license costs, runs to completion on a commodity CPU without a GPU, and adds effectively zero marginal cost per study, which is advantageous for low-resource screening that the WHO identifies as the primary setting for TB CAD without radiologist readers [2].

VII. DISCUSSION

A. Limitations

1) *Limited TB CXR Samples:* TB-specific samples are limited in public datasets. As of writing, TBX11K is the largest public CXR dataset for tuberculosis with labeled bounding boxes, yet its TB subset remains small at 800 TB images and 1,211 TB lesion boxes. While general CXR datasets exist, TB-positive samples are few and insufficient to train from scratch. No public dataset currently contains TB-specific CXRs with reference radiographic reports, which is what an evaluation of clinical correctness would require. This scarcity constrained the development and evaluation of report generation for clinical correctness.

2) *Evaluation on High-burden Populations:* The TB data scarcity limits how representative the evaluation population is. TBX11K is drawn from a single country, from patients presenting for hospital chest examination, and records no HIV status or prior-TB history. The WHO 2025 evaluation instead spans multiple regions, weighted toward high-burden African and South-East Asian settings, includes CXRs from community screening rather than facility screening, and reports lower CAD accuracy among women, people living with HIV, and those with a history of TB [7]. Reported performance here may therefore overstate what the pipeline would achieve in the higher-burden, HIV-prevalent populations that community active case finding targets. Section III-F measures one facet of this gap directly: zero-shot transfer to two other-source sets drops mean AUC_{TB} from 0.998 to 0.564, and although label-free site adaptation recovers part of it, the corrected scores stay below the in-distribution figures. Evaluation on higher-burden, HIV-prevalent populations remains future work.

3) *Evaluation of Clinical Correctness:* Although report generation is faithful by design, faithfulness to the detector is not clinical correctness. A report that faithfully restates an incorrect detection is still wrong, and only reference-based or reader evaluation can report this. However, due to the lack of a TB-specific dataset with reference radiological reports, this is left to future work.

4) *Differences in Pediatric Presentation of TB:* The system's scope is limited to screening of adult patients because the radiographic presentation of pulmonary TB differs in pediatric patients. Adult reactivation tuberculosis appears in the upper lung zones, whereas pediatric primary tuberculosis appears in the lower lung zones. This is consistent with the WHO endorsement of CAD as a human-reader alternative only for patients aged 15 years and older [6].

B. Future Work

1) *Evaluation on Field Hardware:* Future work seeks to extend the commodity-CPU benchmark reported in Table VI to the target field hardware, measuring inference latency and memory on constrained single-board or ARM devices under real-world use.

2) *Detecting Full Radiographic Findings:* Detection of full radiographic findings, such as cardiac size, pleural fluid, or non-TB abnormalities, as well as cardiac, pleural, and skeletal structures, is reserved for future work given additional data. Specific findings are more clinically meaningful for interpretation of the radiologic report, aligned with human reader interpretation of findings. This would also warrant use of a generative language model over a structured template for report generation, because synthesizing findings into a coherent report is no longer a fixed-format substitution but a genuine generation problem. This is demonstrated by recent chest X-ray systems that aggregate region-level findings into a report with an LLM [11], [58].

3) *Reader Study of Generated Reports:* A reader study with radiologists and non-radiologist screeners is also recommended to evaluate the clinical correctness and coherence of the model-generated reports, particularly in comparison to human reader reports.

VIII. CONCLUSION

TB CAD systems are highly accurate, however challenges remain in integrating them for community ACF with unreliable connectivity, limited compute, and no radiologist present. This paper presented a lightweight TB screening pipeline for resource-constrained community ACF settings. The selected FlipR-YOLOv8n configuration completes a study in about 370 ms on this CPU without a GPU. Because the pipeline is open-source, there is no per-study license and effectively zero marginal cost per study, as compared to commercial TB CAD systems with licensing and imaging hardware costs. For localization, a comparison of lightweight detectors covers both active and latent TB, the latter underdetected in prior work. Across all detectors, localization functions as a high-precision rule-in cue rather than a standalone detector. Among the classifiers, FlipR achieves similar accuracy and F1 to the strongest baselines with a parameter-efficient design and the second-highest zero-shot transfer to out-of-distribution data. However, zero-shot mean TB AUC falls from 0.998 to 0.564 on out-of-distribution data. Label-free site adaptation recovers model performance only slightly while staying below the in-distribution and WHO operating targets for TB CAD. The models are recommended to be trained on further data representative of high-risk populations and calibrate per community site as per WHO guidelines. TB-specific samples were found to be highly limited in public datasets, which constrained the design and evaluation of components. Evaluation of deployment feasibility on field hardware, extended radiographic findings for a full radiologist report, and a reader study with radiologists and radiographers is left to future work.

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DECLARATION ON GENERATIVE AI

Generative AI tools were used to assist in writing the manuscript and source code for the study. All generative AI outputs were reverified by the authors. The conception of the research, methodology, and conclusions are the work of the authors.

DATA AND CODE AVAILABILITY

The manuscript, data, and source code for this study is publicly available at <https://github.com/alive-ateneo/tb-community-screening-pipeline>.

APPENDIX A

ARCHITECTURES OF CUSTOM MODELS

APPENDIX B

DOMAIN GAP TO THE EXTERNAL SITES

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TABLE VII. FLIPR ARCHITECTURE (RESNET-18 TRUNCATED TO THREE STAGES)

Stage	Output	Params
Stem (conv 7×7 , BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
FlipR gate	$128 \times H/8$	129
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
Head (avg-pool, FC $256 \rightarrow 3$)	3	771
Total		2,783,684

TABLE VIII. DRAXNET ARCHITECTURE (RESNET-18, DRAXBLOCK IN STAGE 4)

Stage	Output	Params
Stem (conv 7×7 , BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
layer4 (2 DraxBlocks, 512)	$512 \times H/32$	13,699,072
Head (avg-pool, FC $512 \rightarrow 3$)	3	1,539
Total		16,483,395

TABLE IX. DRAX-MOBILENETV3-LARGE ARCHITECTURE

Stage	Output	Params
MobileNetV3-Large features	$960 \times H/32$	2,971,952
DraxBlock adapter (bottleneck 160)	$960 \times H/32$	575,200
Head (avg-pool, classifier $960 \rightarrow 1280 \rightarrow 3$)	3	1,233,923
Total		4,781,075

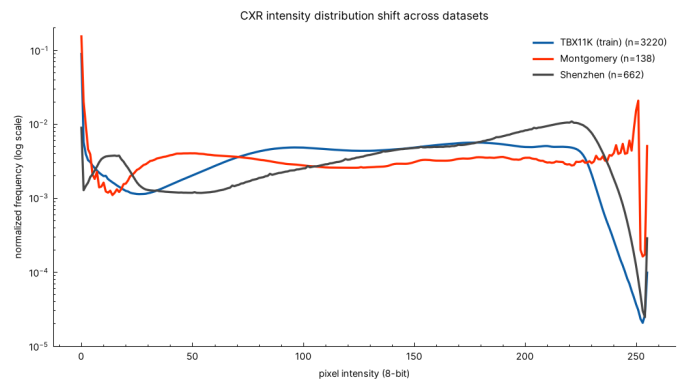


Fig. 6. Pixel-intensity distributions of the TBX11K training images and the two external sites. Montgomery is darker and Shenzhen brighter than the training set, the covariate shift in image appearance discussed in Section III-F that tracks the direction of the zero-shot failure on each site.

[//www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2024](https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2024)

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